IGNACIO FERNÁNDEZ GRAÑA

@ inafergra@gmail.com +34 674762368 Delft, The Netherlands inafergra

Physicist interested in Quantum Information Science, Machine Learning and Computational Methods. Currently investigating the simulation of quantum chemistry with quantum computers. Motivated and self-driven learner, able to quickly adapt to different environments.

EDUCATION

MSc Applied Physics

TU Delft

Sep 2020 - Present Delft, The Netherlands

- Quantum Computing and Quantum Devices track, covering a wide range of topics in quantum information science, quantum technologies, machine learning and computational methods.
- Part of the Honours Programme Delft.
- MSc thesis: *Digital and analog quantum simulation of quantum chemistry* (ongoing).

BSc Physics

University of Santiago de Compostela

Sep 2016 - Jul 2020 Santiago de Compostela, Spain

- 4 year bachelor with a broad spectrum within physics, including courses both in experimental and theoretical/mathematical physics.
- BSc thesis: *Deep Learning for Particle Recognition* at the KVI center in Groningen, The Netherlands. Studied the use of Convolutional Neural Networks for particle tracking in particle physics experiments.
- Year abroad within the Erasmus Programme at University of Groningen, The Netherlands.

WORK EXPERIENCE

Teaching Assistant - Quantum Sensing and Measurement

TU Delft

Sep 2021 - Jan 2022 Delft, The Netherlands

Responsibilities:

- Preparing homework assignments and correcting papers and projects.
- Mentoring the students in the preparation of the course material and the final project.

CERTIFICATIONS

- Deep learning specialization in Coursera (4 online courses)
- Participated in 8 international Erasmus+ Youth Meetings

SKILLS

- Languages: Python, Julia, Matlab, basic knowledge of C++
- Tools: Git, Linux, Latex
- Frameworks: Keras, Scikit-learn, Qiskit, Qutip

PROJECTS

Approximating ground states with free-fermionic states on a quantum computer (Jan 2021 - Jan 2022)

- Honours Programme Research Project at the Applied Quantum Algorithms group (Leiden University).

Percolation theory-based epidemic simulation (Apr 2021 - Jun 2021)

- Simulation of the spread of an epidemic using percolation theory models.

Sparsity as a regularization technique for Quantum Generative Models (Mar 2021 - Jun 2021)

- Study of the effect of sparsity as a regularization technique for Quantum Circuit Born Machines with Cirq.

Monte-Carlo simulation of the Ising model (March 2021 - Apr 2021)

- Monte-Carlo simulation of the 2D Ising model via the Metropolis and Wolff algorithms.

Molecular dynamics simulation of Argon atoms (Feb 2021 - Mar 2021)

- Simulation of a molecular system interacting via Lennard-Jones potential.

Quantum Approximate Optimization Algorithm for the Max-Cut problem (Oct 2020 - Jan 2021)

- Implemented QAOA from scratch using Qiskit.

LANGUAGES

- English (Professional working proficiency)
- Spanish (Native)
- Galician (Native)

PERSONAL INTERESTS

- Music: multi-instrumentalist, music theory
- Sports: handball, boxing, climbing
- Puzzles: Rubik's cubes